Dynamic Response 2

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Reference book for the presentation: **Feedback Control of Dynamic Systems, 8th Edition, Franklin • Powell • Emami-Naeini, Pearson, 2019, ISBN: 978-0-13-468571-7**

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6. Poles and Zeros
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1. One-sided Laplace transform

L_ Laplace transform of f (t):

- Built-in convergence factor if $\sigma_1 > 0$
- If f(t) does not $\rightarrow 0$ as $t \rightarrow \infty$ => the integrand will vanish for sufficiently large values of σ (if f(t) does not grow at a faster-than-exponential rate)
- The lower limit of integration is at 0^- allows the use of an impulse at t = 0
- **L** symbol from here on means **L_**

$$\mathcal{L}_-\{f(t)\} = F(s)$$

$$F(s) \triangleq \int_{0^-}^{\infty} f(t)e^{-st}\,dt.$$

$$s = \sigma_1 + j\omega.$$

- Inverse of the **L_** transform (see equation) where σ_c is to the right of all the singularities of F(s) in the s-plane
- This integral is seldom used => partial-fraction expansion is more commonly used

$$f(t) = \frac{1}{2\pi j}\int_{\sigma_c - j\infty}^{\sigma_c + j\infty} F(s)e^{st}\,ds$$

1. One-sided Laplace transform

Table of Laplace Transforms

Number	$F(s)$	$f(t), t \geq 0$
1	1	$\delta(t)$
2	$1/s$	$1(t)$
3	$1/s^2$	t
4	$2!/s^3$	t^2
5	$3!/s^4$	t^3
6	$m!/s^{m+1}$	t^m
7	$\frac{1}{s+a}$	e^{-at}
8	$\frac{1}{(s+a)^2}$	te^{-at}
9	$\frac{1}{(s+a)^3}$	$\frac{1}{2!}t^2e^{-at}$
10	$\frac{1}{(s+a)^m}$	$\frac{1}{(m-1)!}t^{m-1}e^{-at}$
11	$\frac{a}{s(s+a)}$	$1-e^{-at}$
12	$\frac{a}{s^2(s+a)}$	$\frac{1}{a}(at-1+e^{-at})$
13	$\frac{b-a}{(s+a)(s+b)}$	$e^{-at}-e^{-bt}$
14	$\frac{s}{(s+a)^2}$	$(1-at)e^{-at}$
15	$\frac{a^2}{s(s+a)^2}$	$1-e^{-at}(1+at)$
16	$\frac{(b-a)s}{(s+a)(s+b)}$	$be^{-bt}-ae^{-at}$
17	$\frac{a}{s^2+a^2}$	$\sin at$
18	$\frac{s}{s^2+a^2}$	$\cos at$
19	$\frac{s+a}{(s+a)^2+b^2}$	$e^{-at}\cos bt$
20	$\frac{b}{(s+a)^2+b^2}$	$e^{-at}\sin bt$
21	$\frac{a^2+b^2}{s[(s+a)^2+b^2]}$	$1-e^{-at}\left(\cos bt+\frac{a}{b}\sin bt\right)$

2. Properties of Laplace transform

Properties of Laplace Transforms

Number	Laplace Transform	Time Function	Comment
—	$F(s)$	$f(t)$	Transform pair
1	$\alpha F_1(s) + \beta F_2(s)$	$\alpha f_1(t) + \beta f_2(t)$	Superposition
2	$F(s)e^{-s\lambda}$	$f(t-\lambda)$	Time delay ($\lambda \geq 0$)
3	$\frac{1}{\lvert a\rvert}F\left(\frac{s}{a}\right)$	$f(at)$	Time scaling
4	$F(s+a)$	$e^{-at}f(t)$	Shift in frequency
5	$s^m F(s) - s^{m-1}f(0)$ $-s^{m-2}\dot{f}(0) - \cdots - f^{(m-1)}(0)$	$f^{(m)}(t)$	Differentiation

2. Properties of Laplace transform

6	$\frac{1}{s}F(s)$	$\int_0^t f(\xi)\,d\xi$	Integration
7	$F_1(s)F_2(s)$	$f_1(t) * f_2(t)$	Convolution
8	$\lim_{s\to\infty} sF(s)$	$f(0^+)$	Initial Value Theorem
9	$\lim_{s\to 0} sF(s)$	$\lim_{t\to\infty} f(t)$	Final Value Theorem
10	$\frac{1}{2\pi j}\int_{\sigma_c-j\infty}^{\sigma_c+j\infty} F_1(\xi)F_2(s-\xi)d\xi$	$f_1(t)f_2(t)$	Time product
11	$\frac{1}{2\pi}\int_{-j\infty}^{+j\infty} Y(-j\omega)U(j\omega)\,d\omega$	$\int_0^\infty y(t)u(t)\,dt$	Parseval's Theorem
12	$-\frac{d}{ds}F(s)$	$tf(t)$	Multiplication by time

3. Inverse Laplace transform by partial-fraction expansion

- F(s) of any physical system (m ≤ n)
- Zeros: s = z_i (values that make F(s) = 0)
- Poles: s = p_i (values that make F(s) = ∞)
- Assuming the poles {p_i} are **real** or **complex** but **distinc**

$$F(s) = \frac{b_1 s^m + b_2 s^{m-1} + \cdots + b_{m+1}}{s^n + a_1 s^{n-1} + \cdots + a_n}$$

$$F(s) = K \frac{\Pi_{i=1}^{m}(s - z_i)}{\Pi_{i=1}^{n}(s - p_i)}.$$

$$F(s) = \frac{C_1}{s - p_1} + \frac{C_2}{s - p_2} + \cdots + \frac{C_n}{s - p_n}$$

3. Inverse Laplace transform by partial-fraction expansion

- Cover-up method to calculate the constants:
  - Multiply both sides by the factor $(s-p_1)$
  - Evaluate the left side for p_1
- To calculate C_i
- We know that F(s) ⇔ f(t) for a real pole
- Then f(t) from F(s)

$$F(s) = \frac{C_1}{s-p_1} + \frac{C_2}{s-p_2} + \cdots + \frac{C_n}{s-p_n}$$

$$(s-p_1)F(s) = C_1 + \frac{s-p_1}{s-p_2}C_2 + \cdots + \frac{(s-p_1)}{s-p_n}C_n$$

$$C_1 = (s-p_1)F(s)|_{s=p_1}$$

$$C_i = (s-p_i)F(s)|_{s=p_i}$$

$$F(s) = \frac{1}{s-p_i} \qquad f(t) = e^{p_i t}1(t)$$

$$f(t) = \sum_{i=1}^{n} C_i e^{p_i t}1(t)$$

3. Inverse Laplace transform by partial-fraction expansion

Example: calculate y(t) from Y(s)

$$Y(s) = \frac{(s+4)(s+3)}{s(s+9)(s+2)}$$

3. Inverse Laplace transform by partial-fraction expansion

Example: calculate y(t) from Y(s)

$$Y(s) = \frac{(s+4)(s+3)}{s(s+9)(s+2)}$$

$$Y(s) = \frac{C_1}{s} + \frac{C_2}{s+9} + \frac{C_3}{s+2}$$

$$C_1 = \left.\frac{(s+4)(s+3)}{(s+9)(s+2)}\right|_{s=0} = \frac{2}{3}$$

$$C_2 = \left.\frac{(s+4)(s+3)}{s(s+2)}\right|_{s=-9} = \frac{10}{21}$$

$$C_3 = \left.\frac{(s+4)(s+3)}{s(s+9)}\right|_{s=-2} = -\frac{1}{7}$$

$$y(t) = \frac{2}{3}1(t) + \frac{10}{21}e^{-9t}1(t) - \frac{1}{7}e^{-2t}1(t)$$

Verification with Matlab:

```
num = conv([1 4],[1 3]);        % form numerator polynomial
den = conv([1 9 0],[1 2]);      % form denominator polynomial
[r,p,k] = residue(num,den);     % compute the residues
```

```
r = [0.4762 -0.1429 0.6667]';    p = [-9 -2 0]';    k = [];
```

4. The final value theorem

- To calculate the constant steady-state (t -> ∞) value of y(t) given its Laplace transform Y(s)
- 3 Options for the “limit”
  - Constant
  - Undefined
  - Unbounded
- If any pole of F(s) has a real part > 0 ($Re(p_i) > 0$) => y(t) will be unbounded
- If F(s) has a pair of poles on the imaginary axis of the s-plane ($p_i = \pm j\omega$) => y(t) will contain a sinusoid that persists forever
- If all poles of Y (s) are in the left half of the s-plane (except for one at s = 0) => all terms of y(t) will decay to zero (except the term corresponding to the pole at s = 0 => constraint in time)

4. The final value theorem

If all poles of $sY(s)$ are in the left half of the s-plane, then

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s).$$

4. The final value theorem

Example 1: calculate the final value for the output of a system described by F(s)

$$Y(s) = \frac{7(s+5)}{2s(s^2+2s+3)}$$

Example 2: calculate the final value for the output of a system described by F(s)

$$Y(s) = \frac{4}{s(s^2-5s+6)}$$

4. The final value theorem

Example 1: calculate the final value for the output of a system described by F(s)

$$Y(s) = \frac{7(s+5)}{2s(s^2+2s+3)}$$

$$y(\infty) = sY(s)|_{s=0} = \frac{7 \cdot 5}{2 \cdot 3} = 5.83$$

Example 2: calculate the final value for the output of a system described by F(s)

$$Y(s) = \frac{4}{s(s^2-5s+6)}$$

There are 2 poles with positive real parts

~~$$y(\infty) = sY(s)|_{s=0} = \frac{2}{3}$$~~

$$y(t) = \frac{2}{3} - 2e^{2t} + \frac{4}{3}e^{3t}$$

4. The final value theorem

This theorem can also be used to find the DC gain:

- For a system described by a tf G(s):
  - DC gain is the final value for U(s) = 1/s (unitary step)

$$\text{DC gain} = \lim_{s \to 0} sG(s)\frac{1}{s} = \lim_{s \to 0} G(s)$$

4. The final value theorem

Example: calculate the value of **A** for a system described by G(s) to obtain a DC gain of 2.5

$$G(s) = \frac{4(s + A)}{s^2 + 7s + 5}$$

4. The final value theorem

Example: calculate the value of **A** for a system described by G(s) to obtain a DC gain of 2.5

$$G(s) = \frac{4(s + A)}{s^2 + 7s + 5}$$

$$\text{DC gain} = G(s)|_{s=0} = \frac{4A}{5} = 2.5.$$

$$A = 3.125$$

5. Using Laplace transform to solve differential equations

Example: Homogeneous Differential Equation **(initial conditions different than 0)**

$$\ddot{y}(t) + y(t) = 0, \qquad \text{where} \quad y(0) = \alpha,\ \dot{y}(0) = \beta.$$

5. Using Laplace transform to solve differential equations

Example: Homogeneous Differential Equation **(initial conditions different than 0)**

$$\ddot{y}(t) + y(t) = 0, \qquad \text{where} \quad y(0) = \alpha,\ \dot{y}(0) = \beta.$$

$$\mathcal{L}\{\ddot{f}\} = s^2 F(s) - sf(0^-) - \dot{f}(0^-)$$ Using the ”derivative” property

$$s^2 Y(s) - \alpha s - \beta + Y(s) = 0,$$

$$(s^2 + 1) Y(s) = \alpha s + \beta,$$

$$Y(s) = \frac{\alpha s}{s^2 + 1} + \frac{\beta}{s^2 + 1}$$

$$y(t) = [\alpha \cos t + \beta \sin t]\, 1(t)$$

5. Using Laplace transform to solve differential equations

Example: Forced Differential Equation **(initial conditions different than 0)**

$$\ddot{y}(t) + 5\dot{y}(t) + 4y(t) = 3,$$

where $y(0) = \alpha$, $\dot{y}(0) = \beta$.

5. Using Laplace transform to solve differential equations

Example: Forced Differential Equation **(initial conditions different than 0)**

$$\ddot{y}(t) + 5\dot{y}(t) + 4y(t) = 3,$$

where $y(0) = \alpha,\ \dot{y}(0) = \beta.$

$$s^2 Y(s) - s\alpha - \beta + 5\left[sY(s) - \alpha\right] + 4Y(s) = \frac{3}{s}$$

$$Y(s) = \frac{s(s\alpha + \beta + 5\alpha) + 3}{s(s+1)(s+4)}.$$

$$Y(s) = \frac{\frac{3}{4}}{s} - \frac{\frac{3-\beta-4\alpha}{3}}{s+1} + \frac{\frac{3-4\alpha-4\beta}{12}}{s+4}$$

$$y(t) = \left(\frac{3}{4} + \frac{-3+\beta+4\alpha}{3}e^{-t} + \frac{3-4\alpha-4\beta}{12}e^{-4t}\right)1(t).$$

It can be verified by differentiating this solution twice and substituting the result in the original differential equation

5. Using Laplace transform to solve differential equations

Example: Forced Differential Equation **(0 initial conditions)**

! Simpler !

- With partial-fraction
- Matlab

$$\dot{y}(t) + 6y(t) = u(t),\ y(0) = 0,\ u(t) = 1 - 0.3e^{-3t}$$

$$sY(s) + 6Y(s) = \frac{1}{s} - \frac{0.3}{s+3}$$

$$Y(s) = \frac{\frac{1}{6}}{s} + \frac{\frac{1}{10}}{s+3} + \frac{\frac{2}{30}}{s+6}$$

$$Y(s) = \frac{0.7s + 3}{s(s+3)(s+6)}$$

$$y(t) = \left(\frac{1}{6} - \frac{1}{10}e^{-3t} - \frac{2}{30}e^{-6t}\right) 1(t)$$

5. Using Laplace transform to solve differential equations

Example: Forced Differential Equation **(0 initial conditions)**

- With partial-fraction

$$\dot{y}(t) + 6y(t) = u(t),\ y(0) = 0,\ u(t) = 1 - 0.3e^{-3t}$$

$$sY(s) + 6Y(s) = \frac{1}{s} - \frac{0.3}{s+3}$$

$$Y(s) = \frac{0.7s + 3}{s(s+3)(s+6)}$$

$$Y(s) = \frac{\frac{1}{6}}{s} + \frac{\frac{1}{10}}{s+3} + \frac{\frac{2}{30}}{s+6}$$

$$y(t) = \left(\frac{1}{6} - \frac{1}{10}e^{-3t} - \frac{2}{30}e^{-6t}\right)1(t)$$

5. Using Laplace transform to solve differential equations

Example: Forced Differential Equation **(0 initial conditions)**

- With partial-fraction

$$\dot{y}(t) + 6y(t) = u(t),\ y(0) = 0,\ u(t) = 1 - 0.3e^{-3t}$$

```
clear; clc;

num = [0.7 3]; % form numerator
den = poly([0;-3;-6]); % form denominator polynomial

[r,p,k] = residue(num,den); % compute the residues
disp(r'); disp(p'); disp(k');
```

```
r' = [-0.0667   -0.1000    0.1667]
p' = [-6    -3     0]
k' = [ ]
```

$$y(t) = \left(\frac{1}{6} - \frac{1}{10}e^{-3t} - \frac{2}{30}e^{-6t}\right)1(t)$$

5. Using Laplace transform to solve differential equations

- With the poles of Y(s) => we know what kind of characteristic terms will appear in the response
- In the last example:
  - The pole at s = –1 => y = Ce^{-t} in the response
  - The pole at s = –4 => y = Ce^{-4t} in the response (decays faster)
  - If there there had been a pole at s = +1 => y = Ce^{+t} (increasing response)
- The poles provide information about the transient response of the system

6. Poles and Zeros

$$H(s) = \frac{b_1 s^m + b_2 s^{m-1} + \cdots + b_{m+1}}{s^n + a_1 s^{n-1} + \cdots + a_n} = \frac{N(s)}{D(s)}$$

- A rational transfer function can be described either as a ratio of two polynomials in s
- K => transfer function gain
- Roots of the numerator => z1, z2, ... , zm are called the **finite zeros**
- Capability to block frequencies coinciding with its zero locations
- The **zeros** have a significant effect on the transient properties

$$H(s) = K \frac{\prod_{i=1}^{m} (s - z_i)}{\prod_{i=1}^{n} (s - p_i)}$$

$$H(s)|_{s=z_i} = 0.$$

$$u = u_0 e^{s_0 t} \qquad s_0 = z_i.$$

$$y \equiv 0$$

6. Poles and Zeros

$$H(s) = \frac{b_1 s^m + b_2 s^{m-1} + \cdots + b_{m+1}}{s^n + a_1 s^{n-1} + \cdots + a_n} = \frac{N(s)}{D(s)}$$

- Roots of the denominator => p1, p2, ... , pn are called the **poles**
- The **poles** determine
  - System's stability properties
  - System's natural or unforced behavior (or the **modes**)
- The system is said to have **n–m** zeros at **infinity** if m < n because H(s) -> 0 as s -> ∞
- If zi = pj => then there are cancellations in the transfer function (undesirable system properties)

$$H(s) = K \frac{\prod_{i=1}^{m} (s - z_i)}{\prod_{i=1}^{n} (s - p_i)}$$

$$|H(s)|_{s=p_i} = \infty.$$

7. Linear system analysis using Matlab

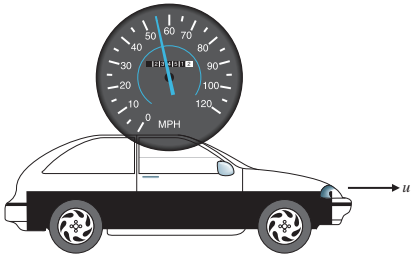
- Step 1: set of time-domain differential equations (representing the dynamic behavior)
- Step 2: to determine and designate inputs and outputs of the system
- Step 3: to compute the transfer function characterizing the input–output behavior of the system

- Analyzing the transfer function we can determine the dynamic properties of the system:
  - Determining the pole–zero locations and deduce the essential characteristics of the dynamic properties
  - Determining the time-domain properties of the system by determining the response to typical excitation: impulses, steps, ramps, and sinusoids.
  - Determining time response analytically (inverse Laplace transform – partial fraction)

7. Linear system analysis using Matlab

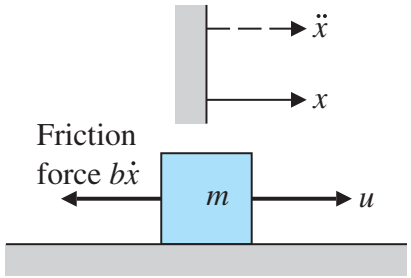
Example 1: Cruise Control Transfer Function Using Matlab

Find the transfer function between the input u and the position of the car x in the cruise control system



$$u - b\dot{x} = m\ddot{x}, \qquad \ddot{x} + \frac{b}{m}\dot{x} = \frac{u}{m}.$$

$$H(s) = \frac{0.001}{s^2 + 0.05s} = \frac{0.001}{s(s + 0.05)}$$




```
s=tf('s');                  % define Laplace variable
sysH=0.001/(s^2+0.05*s);    % form transfer function

p=pole(sysH);               % compute poles
[z,k]=zero(sysH);           % compute zeros and transfer function gain
```

$p = [0 \quad -0.05]'$, and $k = 0.001$

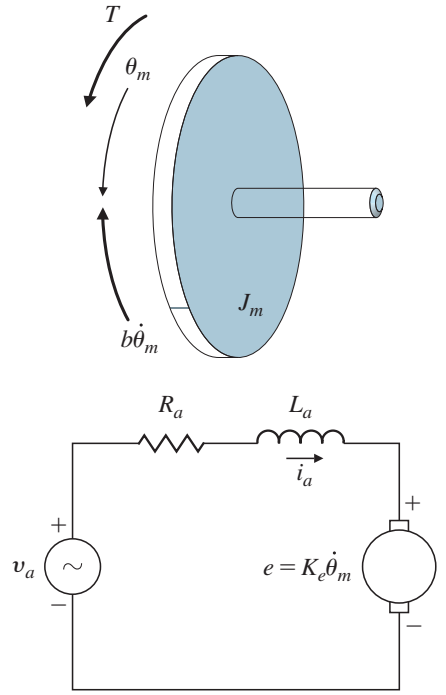
7. Linear system analysis using Matlab

Example 2: DC Motor Transfer Function Using Matlab

For the example explained in a previous lesson: assume that Jm = 0.02 kg·m2, b = 0.005N·m· sec, Kt = Ke = 0.5, Ra = 2.5, and La = 0.1H.

Find the transfer function between the input va and

1. the output θm,
2. the output ωm.



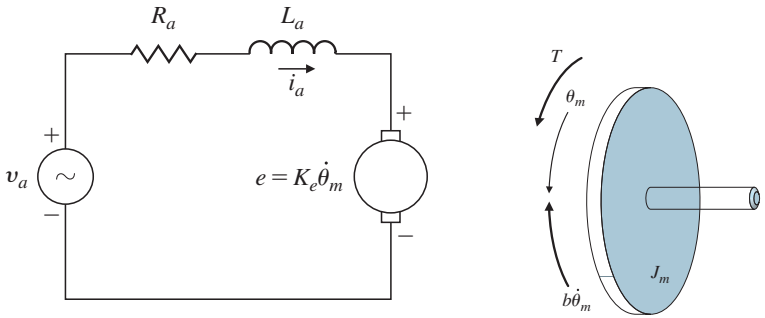
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Find the transfer function between the input va and

1. the output θm,



$$\frac{\Theta_m(s)}{V_a(s)} = \frac{K_t}{s\left[(J_m s + b)(L_a s + R_a) + K_t K_e\right]}$$

$$H(s) = \frac{250}{s^3 + 25.25s^2 + 131.25s} = \frac{250}{s(s^2 + 25.25s^2 + 131.25)}$$

```
s = tf ('s');                                  % define Laplace variable
sysH = 250/(s^3+25.25*s^2+131.25*s);           % form transfer function
```

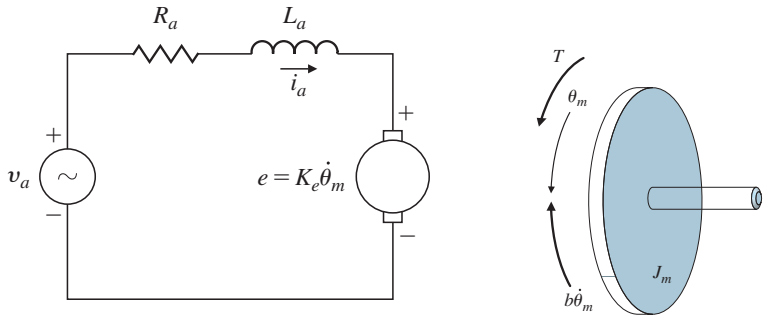
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Find the transfer function between the input va and

1. the output θm,




```
s = tf ('s');                                  % define Laplace variable
sysH = 250/(s^3+25.25*s^2+131.25*s);           % form transfer function

p = pole(sysH);          % compute poles
[z,k] = zero(sysH);      % compute zeros and transfer function gain
```

$$z = [\,],\quad p = [0 \quad -17.9298 \quad -7.3202]',\quad k = 250,$$

$$H(s) = \frac{250}{s(s+17.9298)(s+7.3202)}$$

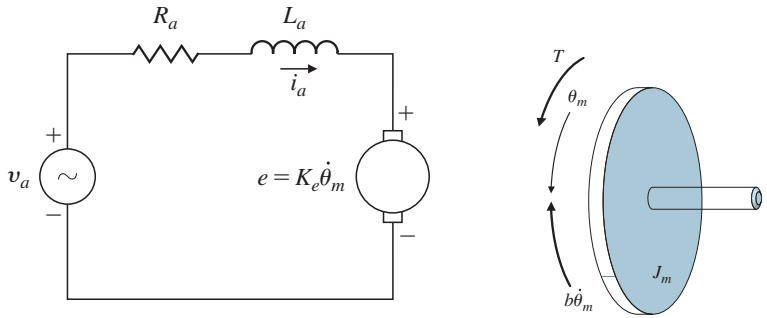
7. Linear system analysis using Matlab

Example 2: DC Motor Transfer Function Using Matlab

For the example explained in a previous lesson: assume that Jm = 0.02 kg·m2, b = 0.005N· m· sec, Kt = Ke = 0.5, Ra = 2.5, and La = 0.1H.

Find the transfer function between the input va and

2. the output ωm.



$$G(s) = \frac{250s}{s^3 + 25.25s^2 + 131.25s} = \frac{250}{s^2 + 25.25s + 131.25}$$

```
s = tf('s');                          % define Laplace variable
sysG = 250/(s^2+25.25*s+131.25);      % form transfer function
t = 0:0.01:4;                         % form time vector
y = step(sysG,t);                     % compute step response;
plot(t,y);                            % plot step response
```

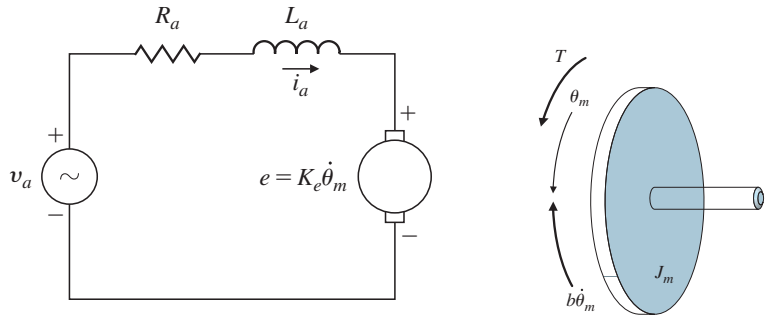
7. Linear system analysis using Matlab

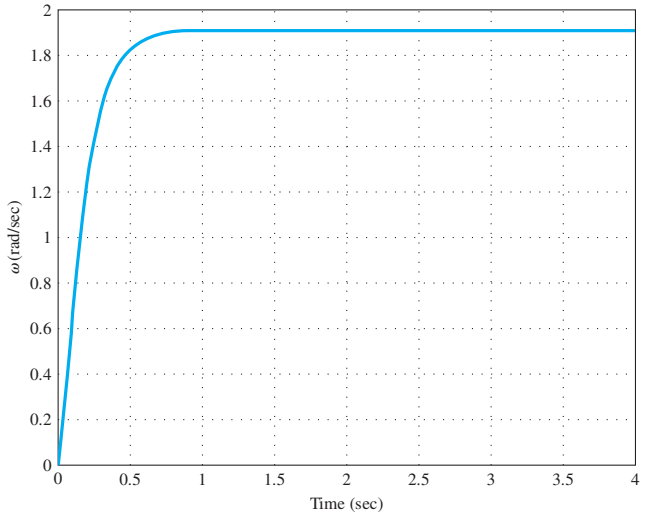
Example 2: DC Motor Transfer Function Using Matlab

For the example explained in a previous lesson: assume that Jm = 0.02 kg·m2, b = 0.005N· m· sec, Kt = Ke = 0.5, Ra = 2.5, and La = 0.1H.

Find the transfer function between the input va and

2. the output ω = θm.



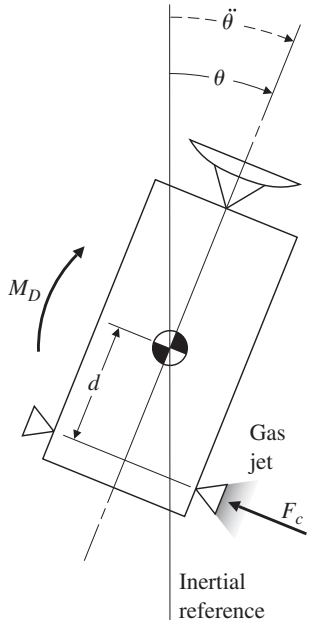


7. Linear system analysis using Matlab

Example 3: Satellite Transfer Function Using Matlab

1. Find the transfer function between the input Fc and the satellite attitude θ in the example of a previous lesson with $M_D = 0$ (PV torque)

Let d= 1 m and I = 5000 kg·m²



$$F_c d + M_D = I\ddot{\theta}.$$

$$\frac{\Theta(s)}{U(s)} = \frac{1}{I}\frac{1}{s^2}$$

$$\frac{d}{I} = \frac{1}{5000} = 0.0002\left[\frac{\text{m}}{\text{kg}\cdot\text{m}^2}\right]$$

$$H(s) = \frac{0.0002}{s^2}$$

7. Linear system analysis using Matlab

Example 3: Satellite Transfer Function Using Matlab

2. Determine the response of the system to a 25-N pulse of 0.1-sec duration, starting at t = 5 sec.

$$H(s) = \frac{0.0002}{s^2}$$

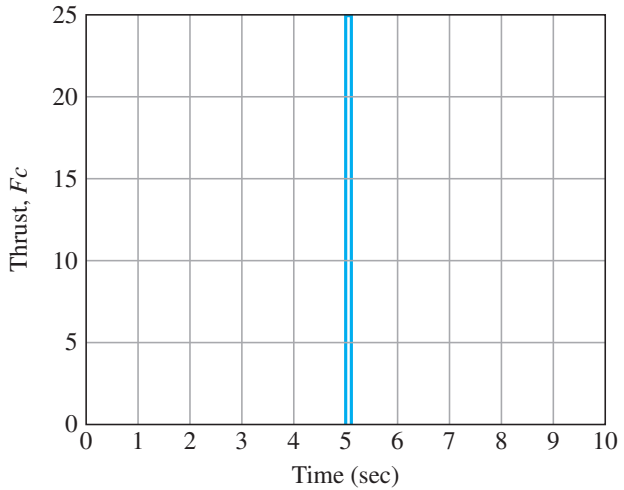
```
clear; clc;
s=tf('s'); % define Laplace variable
sysG=0.0002/s ^2; % define system by its
transfer function
t=0:0.01:10; % set up time vector with dt =
0.01 sec
% pulse of 25N, at 5 sec, for 0.1 sec duration
u1=[zeros(1,500) 25*ones(1,10) zeros(1,491)];
% pulse input
[y1]=lsim(sysG,u1,t); % linear simulation
ff=180/pi; % conversion factor from radians to
degrees
y1=ff*y1; % output in degrees
figure(1); plot(t,u1); % plot input signal
figure(2); plot(t,y1); % plot output response
```

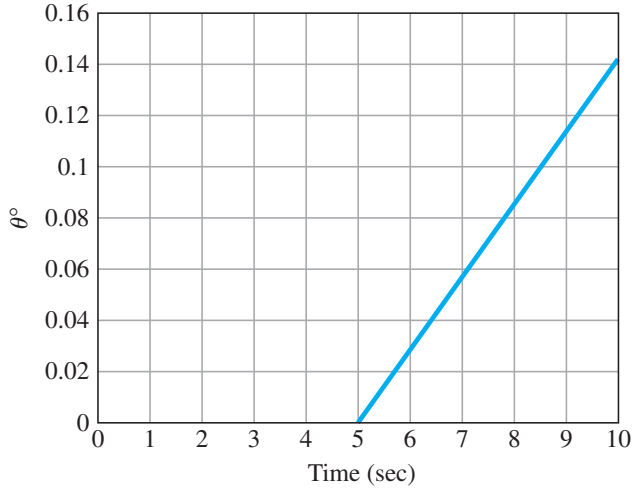
7. Linear system analysis using Matlab

Example 3: Satellite Transfer Function Using Matlab

2. Determine the response of the system to a 25-N pulse of 0.1-sec duration, starting at t = 5 sec.

$$H(s) = \frac{0.0002}{s^2}$$





7. Linear system analysis using Matlab

Example 3: Satellite Transfer Function Using Matlab

2. Determine the response of the system to a 25-N pulse of 0.1-sec, starting at t = 5 sec and a -25-N pulse of 0.1 –sec starting at t = 6.1 sec

$$H(s) = \frac{0.0002}{s^2}$$

```
clear; clc;
s=tf('s'); % define Laplace variable
sysG=0.0002/s ^2; % define system by its
transfer function
t=0:0.01:10; % set up time vector with dt =
0.01 sec
% pulse of 25N, at 5 sec, for 0.1 sec duration
u2=[zeros(1,500) 25*ones(1,10) zeros(1,100) -
25*ones(1,10) zeros(1,381)];
[y2]=lsim(sysG,u2,t); % linear simulation
plot(t,u2); % plot input signal
ff=180/pi; % conversion factor from radians to
degrees
y2=ff*y2; % output in degrees
figure(3); plot(t,y2); % plot output response
```

7. Linear system analysis using Matlab

Example 3: Satellite Transfer Function Using Matlab

2. Determine the response of the system to a 25-N pulse of 0.1-sec, starting at t = 5 sec and a -25-N pulse of 0.1 –sec starting at t = 6.1 sec

$$H(s) = \frac{0.0002}{s^2}$$

